

# Sonal Singh

425-754-2949 | sonalsingh18696@gmail.com | linkedin.com/in/sonalsingh1806 | github.com/sonalsingh1806

AI/ML engineer with 5+ years of production software experience, including hands-on work architecting and deploying agentic AI systems. Builds multi-step reasoning workflows, retrieval and tool-use pipelines, and orchestration logic using LangChain and RAG, with production discipline around evaluation, monitoring, and human-in-the-loop feedback so systems improve with usage. Comfortable across Python-based ML stacks and TypeScript-based production systems, balancing fast iteration with reliability, scalability, and compliance. Available immediately.

## EXPERIENCE

### AI & LLM Engineer (Internship) Future Force — Seattle, WA | June 2025 – Sept 2025

- Architected and shipped production-grade AI agent services in Python and FastAPI, applying microservices architecture and clean-code principles to senior-level technical problems.
- Built and maintained resilient data ingestion and curation pipelines processing structured and unstructured data at scale, feeding downstream agent and retrieval workflows with fault tolerance as a core design requirement.
- Operated as an AI-native engineer, using Cursor and GitHub Copilot to build and orchestrate AI systems, running fast experiment-measure-iterate cycles and increasing shipped output per sprint.
- Reduced end-to-end agent response latency by 40% through systematic performance profiling, caching, and batching, measuring impact at every iteration.
- Wrote unit and integration tests, participated in code reviews, and maintained technical documentation to keep production agent systems reliable and compliant.

### Software Engineer Specialist ABB Limited — Bangalore, India | Oct 2021 – Feb 2024

- Owned backend microservices and REST APIs powering large-scale telemetry data processing in a distributed computing environment, serving as a key point of reliability for the platform.
- Architected schemas and transformation pipelines engineered for performance, reliability, and cost-effectiveness at enterprise scale, reducing rework on downstream consumers.
- Built audit and activity-logging systems from the ground up, closing a visibility gap and enabling faster compliance reporting for the platform team.
- Drove engineering quality as an active code reviewer and led root-cause investigations for production incidents, applying structured RCA to eliminate repeat failures.

### Software Developer Siemens Technology and Services — Gurugram, India | July 2019 – Oct 2021

- Owned the full lifecycle of a Shift Management System, from architecture through deployment and ongoing operations, for high-criticality manufacturing environments.
- Designed and built fault-tolerant REST APIs and service integrations following object-oriented design principles and SDLC best practices.
- Led the migration of a deprecated Silverlight frontend to Angular, mapping components end-to-end to deliver on schedule with a zero-rollback launch.
- Owned CI/CD pipeline configuration in Azure DevOps, including staging validation and deployment automation, and resolved high-pressure production issues by isolating root causes quickly to protect uptime.

## PROJECTS

### SafeSpace AI

Python, TypeScript/React, FastAPI, PostgreSQL, FAISS, LangChain, RAG | Sept 2025 – Present | Live: safespaceai-production.up.railway.app

- Architected and deployed a multi-step AI agent system end to end, orchestrating retrieval, reasoning, and tool-use steps with LangChain, and owning the full lifecycle from design through production deployment and monitoring.
- Built agentic RAG workflows with guardrails and human-in-the-loop review, creating a feedback loop that hardens agent behavior against edge cases and failure modes at runtime.
- Designed and ran an evaluation framework via a real user study (n=13) with statistical analysis of usability and effectiveness, translating findings directly into agent and retrieval-quality iteration — a compounding flywheel for continuous improvement.

### Marketplace Aggregator

AWS Lambda, IAM Roles, API Gateway, DynamoDB, SQS/DLQ, CloudFront | 2024 | Live: d2ctjm6nysi7ht.cloudfront.net

- Architected and deployed a serverless, event-driven system on AWS using Lambda, SQS with dead-letter queues, DynamoDB, and CloudFront.
- Configured IAM least-privilege access policies across all Lambda functions and designed for fault tolerance with asynchronous processing, retry logic, and DLQ-based error handling.

## TECHNICAL SKILLS

**AI/ML & GenAI:** AI Agents & Orchestration, Multi-Step Reasoning, Tool Use, LangChain, RAG, Vector Embeddings, Prompt Engineering, Model Evaluation, Human-in-the-Loop Review, Cursor, Copilot

**Languages:** Python, TypeScript, JavaScript, Java, C#

**Cloud & Infrastructure:** AWS (Lambda, IAM, SQS, DynamoDB, EKS, CloudFront, API Gateway), Azure, Docker, Kubernetes, Terraform, CI/CD, Azure DevOps

**Backend, Data & Systems:** FastAPI, Node.js, RESTful APIs, Microservices, Distributed & Event-Driven Systems, PostgreSQL, DynamoDB, NoSQL, Schema & Data Pipeline Design

**Engineering Practices:** Experimentation & Iteration, OOP, Design Patterns, SDLC, Testing, Code Reviews, Monitoring, Debugging, Agile/CI-CD

## EDUCATION

**M.S. Computer Science and Software Engineering (GPA: 3.87)** University of Washington Bothell | Aug 2024 – June 2026

**B.Tech Software Engineering** SRM University, India | Aug 2015 – May 2019